

Java JDBC Notes

What is JDBC?

JDBC (Java Database Connectivity) is an API for connecting and executing queries with the database from Java programs.

Steps to Use JDBC:

1. Import JDBC packages
2. Load and register the driver
3. Create a connection
4. Create a statement
5. Execute a query
6. Process the results
7. Close the connection

JDBC Driver Example Code:

```
Class.forName("com.mysql.cj.jdbc.Driver");
Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/dbname", "username",
"password");
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery("SELECT * FROM tablename");
while(rs.next()) {
    System.out.println(rs.getInt(1) + " " + rs.getString(2));
}
con.close();
```

Types of JDBC Drivers:

1. Type 1 - JDBC-ODBC Bridge
2. Type 2 - Native-API Driver
3. Type 3 - Network Protocol Driver
4. Type 4 - Thin Driver (Pure Java)

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Common JDBC Interfaces:

- Driver: Interface for all drivers
- Connection: Maintains a session with DB
- Statement: Executes SQL queries
- PreparedStatement: Executes parameterized queries
- ResultSet: Holds data from DB

PreparedStatement Example:

```
PreparedStatement ps = con.prepareStatement("INSERT INTO student VALUES (?, ?)");  
ps.setInt(1, 101);  
ps.setString(2, "John");  
ps.executeUpdate();
```

Exception Handling:

```
try {  
    // JDBC code  
} catch (SQLException e) {  
    e.printStackTrace();  
}
```

JDBC Package:

All JDBC classes are part of java.sql package.

Use Cases:

- Web applications (Servlet/JSP)
- Desktop applications
- Android apps (with JDBC wrappers)